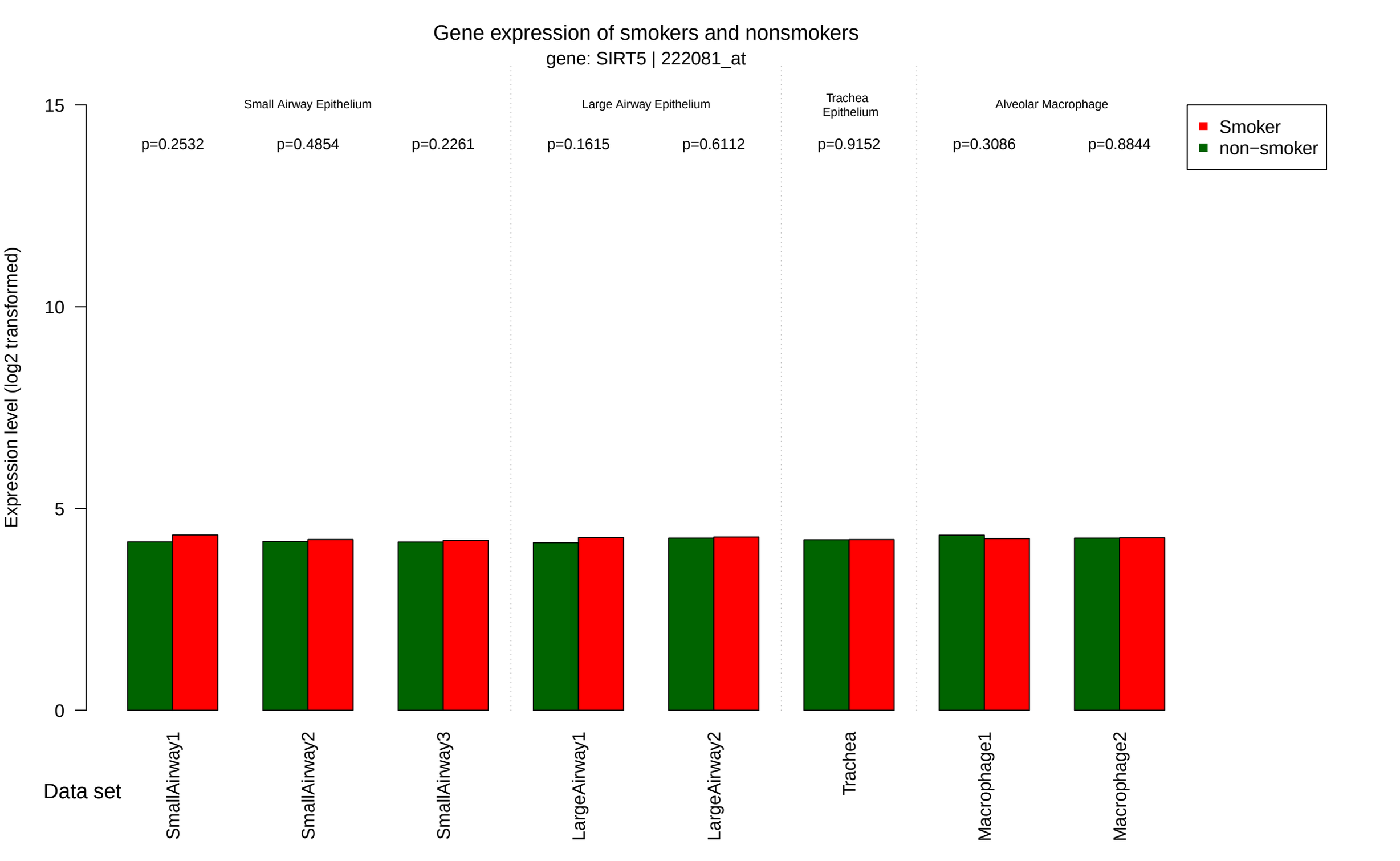
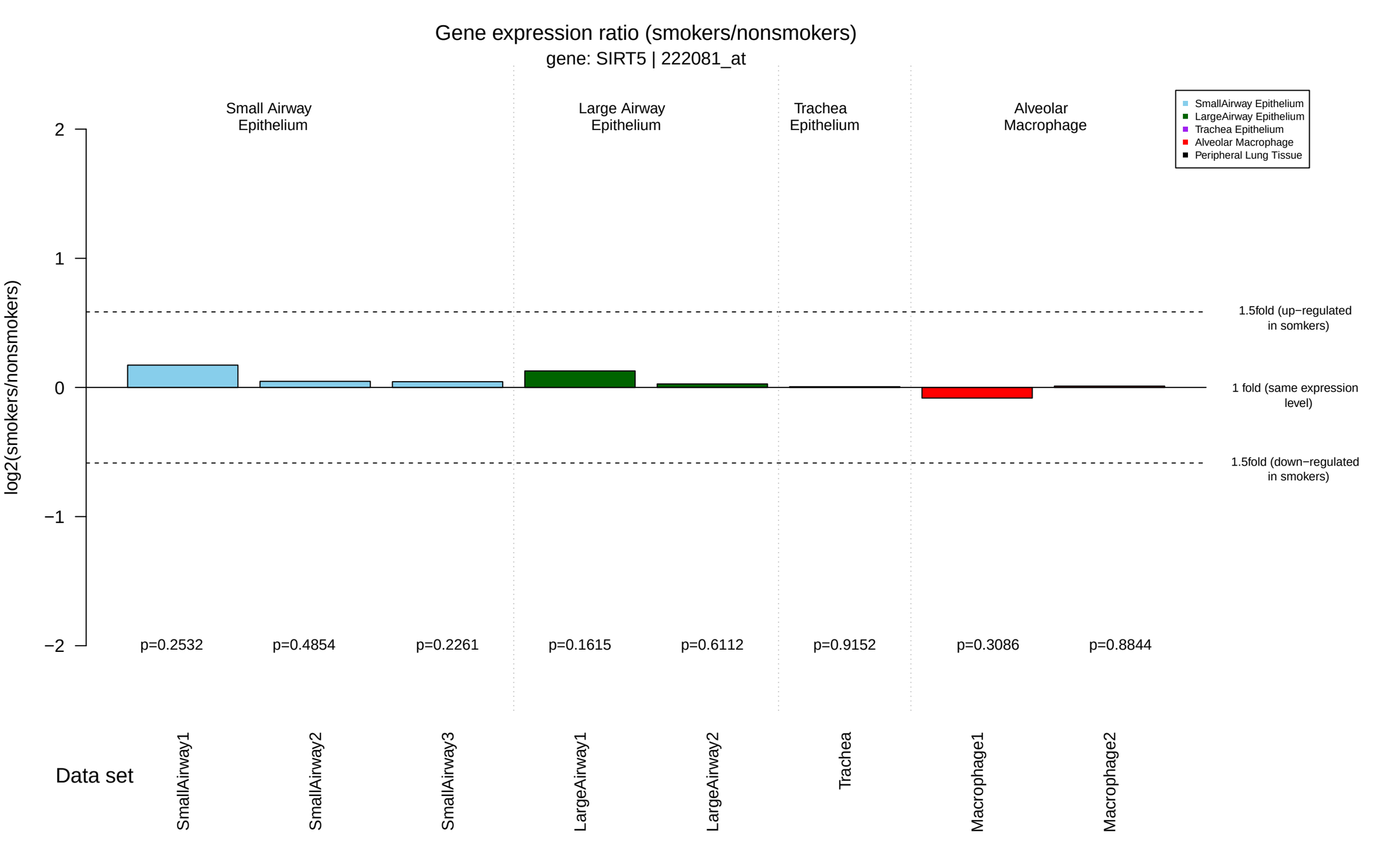
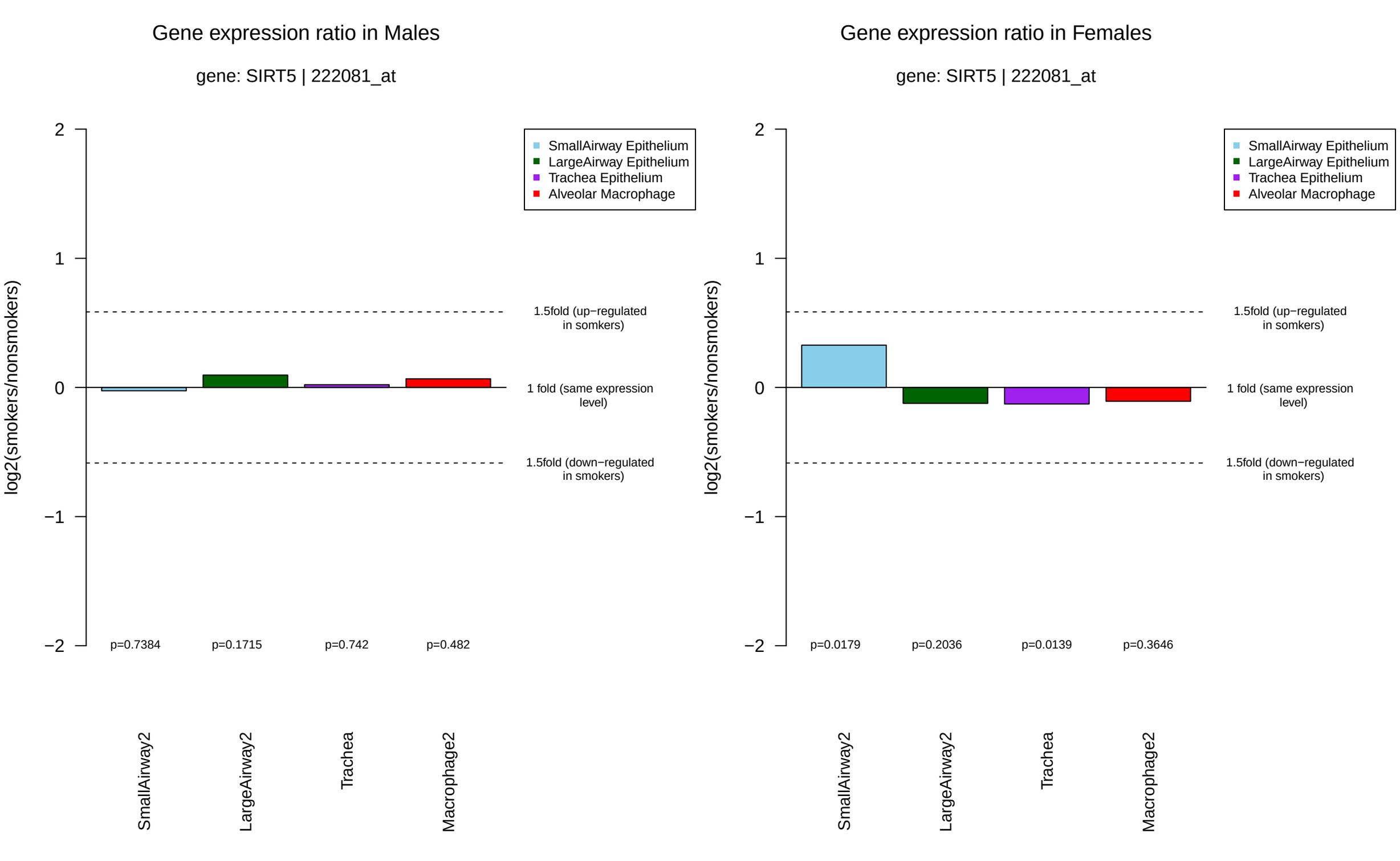
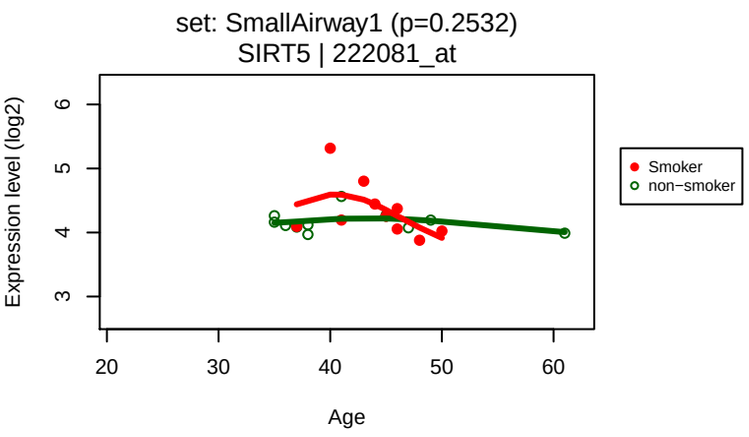
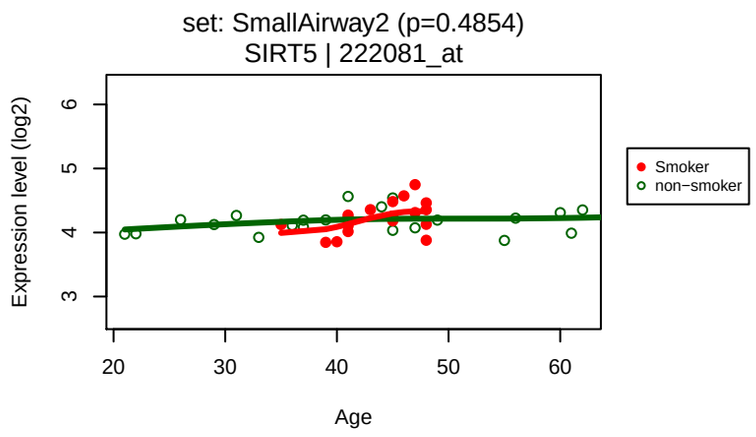
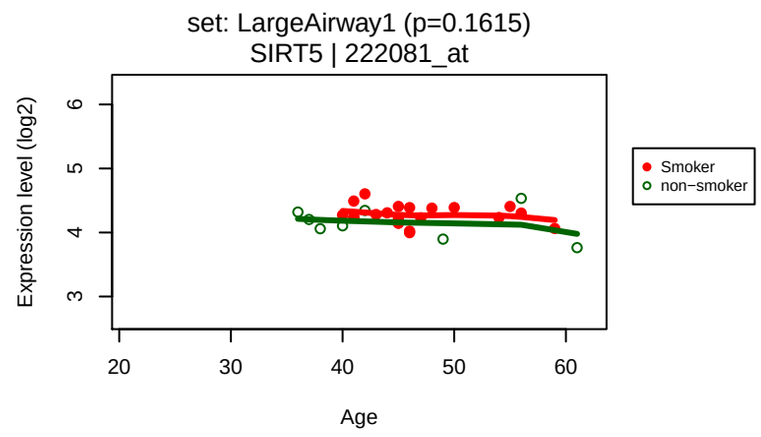

Gene expression of smokers and nonsmokers
gene: SIRT5 | 222081_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2532
p=0.4854
p=0.2261
p=0.1615
p=0.6112
p=0.9152
p=0.3086
p=0.8844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

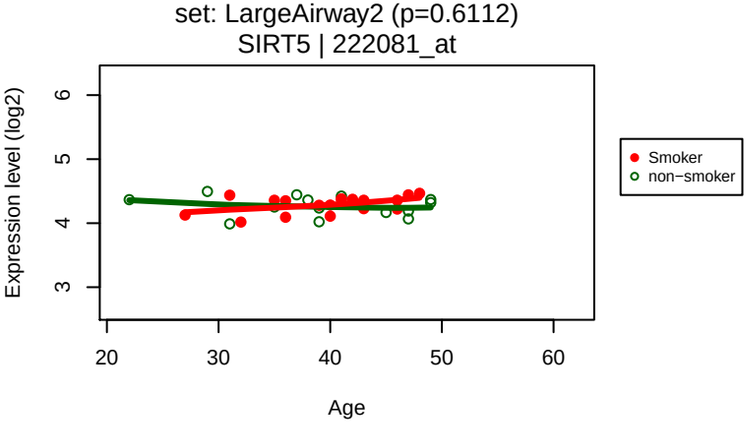

Gene expression ratio (smokers/nonsmokers)
gene: SIRT5 | 222081_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2532
p=0.4854
p=0.2261
p=0.1615
p=0.6112
p=0.9152
p=0.3086
p=0.8844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

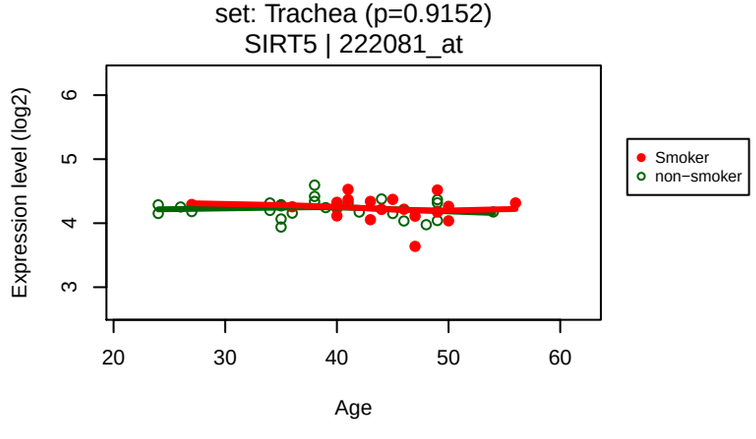

Gene expression ratio in Males
gene: SIRT5 | 222081_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7384
p=0.1715
p=0.742
p=0.482
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT5 | 222081_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0179
p=0.2036
p=0.0139
p=0.3646
SmallAirway2
LargeAirway2
Trachea
Macrophage2

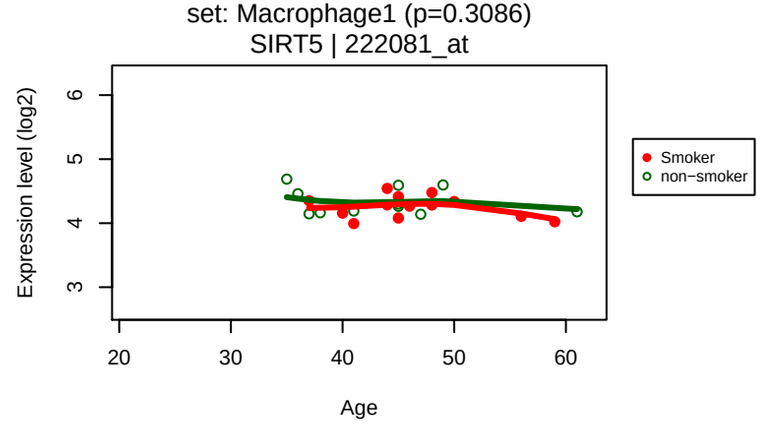

set: SmallAirway1 (p=0.2532)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker

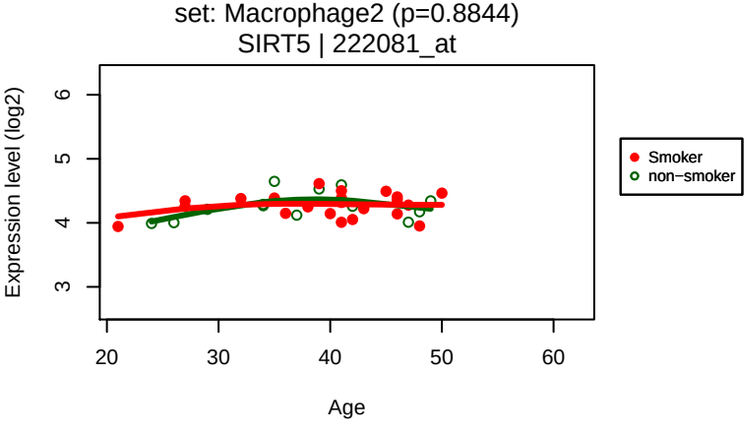

set: SmallAirway2 (p=0.4854)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker

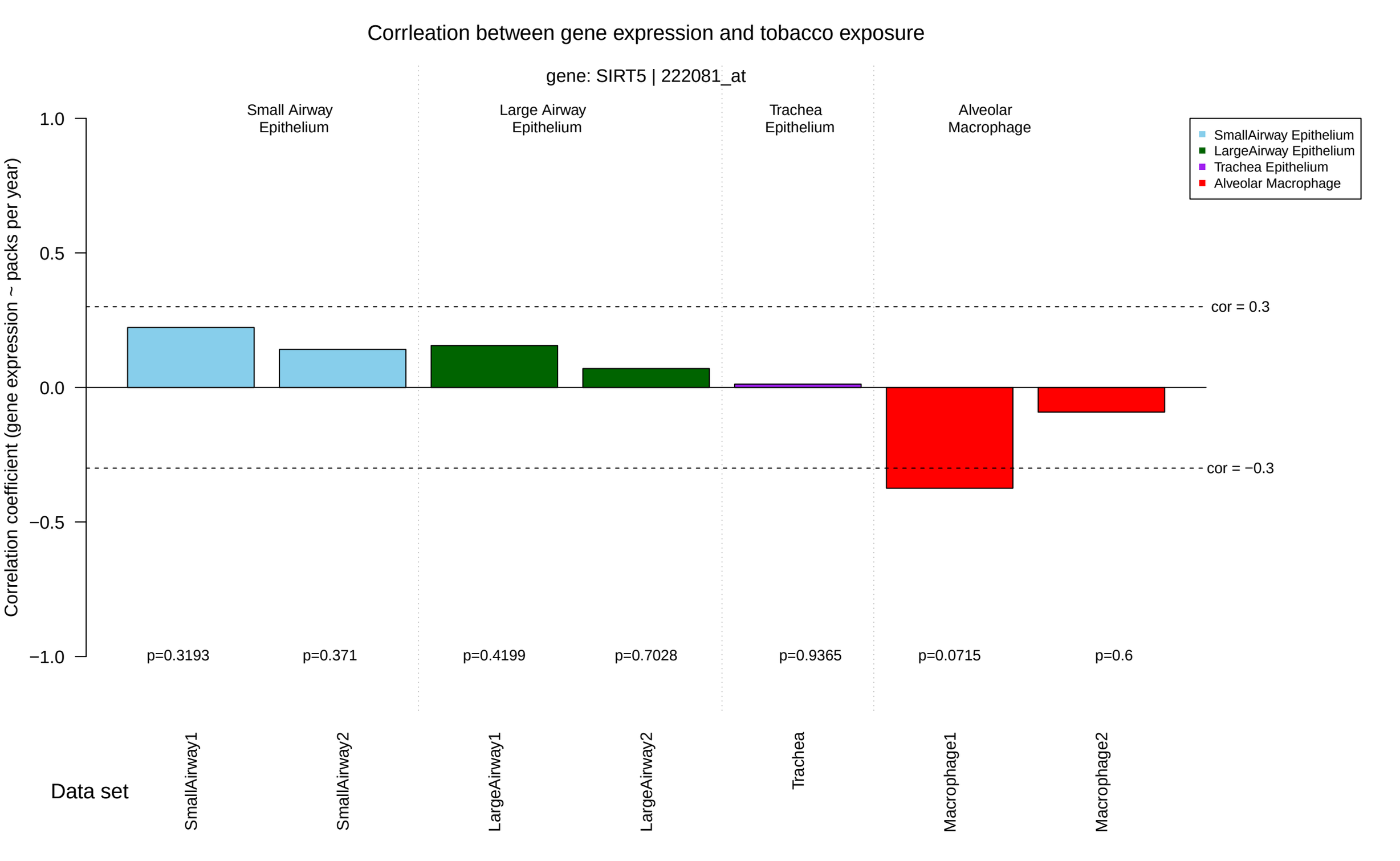

set: LargeAirway1 (p=0.1615)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6112)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9152)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3086)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8844)
SIRT5 | 222081_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT5 | 222081_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3193
p=0.371
p=0.4199
p=0.7028
p=0.9365
p=0.0715
p=0.6
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2